

AAYUSH PATEL

Atlanta, GA | (770) 862-5673 | apatel3088@gatech.edu | U.S. Permanent Resident, no sponsorship required
linkedin.com/in/aysh-ptl | github.com/AayushP-418 | aayush-patel-portfolio-tezw.vercel.app

EDUCATION

Georgia Institute of Technology

B.S. Computer Science, Intelligence and Systems Architecture; Minor: FinTech

Atlanta, GA

Expected May 2028

- GPA: 3.75 / 4.00. Coursework: Data Structures & Algorithms, Computer Organization, Objects & Design, Machine Learning
- Honors: Faculty Honors; Dean's List

TECHNICAL SKILLS

Languages: Go, Java, C++, Python, TypeScript, JavaScript

Systems & Backend: gRPC, REST, WebSockets, Spring Boot, FastAPI, Node.js, Redis, PostgreSQL, Concurrency, Caching

Cloud & Infrastructure: AWS (ECS Fargate, RDS, ElastiCache, SQS, Lambda), GCP, Docker, Terraform, CI/CD, Linux

Testing & Data: k6, Unit/Integration Testing, SQLAlchemy, pgvector/HNSW, ChromaDB, Schema Design, Query Optimization

EXPERIENCE

Big Data Big Impact, Georgia Tech

Jan 2026 - Present

Director of Projects

Atlanta, GA

- Lead 11 AI/ML and software teams and 100+ students shipping production data, NLP, and full-stack systems
- Standardized Git workflows, code review, test gates, and release checklists to keep delivery consistent across teams
- Drove 6 projects from requirements to deployment, presenting architecture and release decisions to GT advisors and Microsoft

Pinnacle Compounding Pharmacy

Aug 2026 - Present

Data Science Intern

Phoenix, AZ (Remote)

- Build Python data services over claims and remittance data on a governed PostgreSQL schema for downstream analytics
- Shipped an LLM extraction pipeline parsing 2K denial letters into structured records at 92% field-level accuracy
- Define leakage-safe validation checks for claim features before they enter modeling and retrieval workflows

Georgia Institute of Technology

May 2026 - Aug 2026

Teaching Assistant, CS 2340: Objects and Design

Atlanta, GA

- Sole TA for a summer section covering Java, OOP, design patterns, Agile/Scrum, and software architecture
- Reviewed class and sequence designs for interface boundaries, testability, and maintainability before implementation
- Guided debugging and refactoring in office hours using SonarQube, helping students turn brittle code into modular designs

Georgia Tech VIP - EMAD Lab (Advisor: Dr. Jason Zutty)

Aug 2025 - May 2026

Undergraduate Researcher, Evolutionary AutoML

Atlanta, GA

- Integrated Gemma, Qwen, Gemini, and DeepSeek-R1 as modular operators in a Python evolutionary search framework
- Engineered an island-migration scheduler across parallel Linux workers, reducing time to convergence by 5.8x
- Ran controlled ablations and system tuning that improved synthesized-solution quality by 14%

PROJECTS

Gatekeeper | github.com/AayushP-418/Gatekeeper

TypeScript | Redis/Lua | gRPC | WebSockets | k6

- Reproduced 34% over-admission at 200 VUs, traced a GET-then-INCR race, and fixed it with atomic Redis Lua
- Added token-bucket limiting, circuit breaking, and live WebSocket telemetry behind the gRPC control plane
- Locked the fix into Dockerized integration and k6 load tests so concurrency regressions fail before release

FeatureFlag | github.com/AayushP-418/Feature-Flag

Go | Redis | AWS SQS | Terraform

- Added Redis write-through caching to Go services, raising same-machine throughput 30% in a controlled local benchmark
- Cut config staleness from 59s to under 1s with SQS invalidation and server-sent updates to SDK clients
- Provisioned ECS Fargate, RDS, ElastiCache, ALB, SQS, and Lambda with Terraform and CI-gated releases

Research Paper RAG | github.com/AayushP-418/research-paper-RAG

Python | FastAPI | PostgreSQL | pgvector/HNSW

- Created a FastAPI backend with durable Postgres conversations, document APIs, and cited vector retrieval
- Enforced Pydantic schemas so downstream services receive predictable, source-grounded model outputs
- Added latency, failure, citation, token, and cost telemetry to separate retrieval from application failures

TerraTrends | github.com/AayushP-418/TerraTrends

Python | Flask | GCP | React

- Integrated 15+ public and cloud data sources into county-level forecasting workflows backed by Flask services
- Connected forecasts to geospatial heatmaps, regional comparison, and competitor-ranking views for decision support

FinLoRA | github.com/AayushP-418/Finlora

Python | PyTorch | Hugging Face | QLoRA | vLLM

- Audited 28.5K synthetic SEC QA pairs, found 82% placeholder leakage, and rebuilt the data pipeline before retraining
- Served the fine-tuned 3.8B model through FastAPI/vLLM after comparing base, SFT, and DPO behavior on held-out data